Problem Sheet 6 Solutions

1.

(a) Averaging the rectified signal over 1 cycle of period T we can write

$$\langle V_{out}\rangle = \frac{1}{T}\left(\int_0^{\frac{T}{2}} A\sin(\omega t+\phi)\,dt - \int_{\frac{T}{2}}^{T} A\sin(\omega t+\phi)\,dt\right)$$

(b) Since

$$\sin(\omega t+\phi) = \sin\omega t\cos\phi + \cos\omega t\sin\phi$$

$$\int \sin(\omega t+\phi)dt = \cos\phi\int \sin\omega t\,dt + \sin\phi\int\cos\omega t\,dt$$

$$\int_0^{T/2}\sin\omega t\,dt = -\int_{T/2}^{T}\sin\omega t\,dt = \frac{2}{\omega}$$

$$\int_0^{T/2}\cos\omega t\,dt = \int_{T/2}^{T}\cos\omega t\,dt = 0$$

$$\omega = \frac{2\pi}{T}$$

$$\langle V_{out}\rangle = \frac{2A\cos\phi}{\pi}.$$

2.

(a) Defining the temperature difference ΔT as

$$\Delta T = T_0 - T,$$

such that it is positive for our case, then we have dQ/dt, the rate of heat transfer from the heat-bath to the mercury, as

$$\frac{dQ}{dt} = \frac{\kappa\Delta T A}{d}.$$

(b) Q is related to the temperature of the mercury T by $Q = mc\Delta T$ and, making the substitution

$$a = \frac{\kappa A}{mcd},$$

we get the 1st order equation

$$\frac{dT}{dt} + aT = aT_0.$$

Multiplying by an integrating factor $e^{\int a dt}$

$$e^{at}\frac{dT}{dt} + e^{at}aT = e^{at}aT_0$$

$$\frac{d}{dt}[e^{at}T] = e^{at}T_0.$$

Integrating using the initial condition $T(0) = 0$ gives

$$T = T_0(1 - e^{-at}) = T_0\left(1 - e^{-\frac{t\kappa A}{mcd}}\right).$$

(c) In the thermal system the temperature difference across the thermal resistance results in a flow of heat (a conserved quantity). The rise in temperature is proportional to the heat capacity of the mercury and the heat it stores. In an electrical system a potential difference across a resistor results in a flow of charge (a conserved quantity). The voltage across a capacitor is proportional to its electrical capacitance and the charge it stores. We can make an equivalence between thermal resistance and electrical resistance, between heat capacity and electrical capacitance, and between temperature and voltage. We can also compare rate of heat flow (dQ/dt) with electrical current (di/dt). The electrical equivalent circuit is charging up a capacitor through a resistor (low-pass filter).